

MPFS Updating Tutorial

This tutorial teaches you how to update MPFS-ICICLE-KIT-ES to latest software version based on my successful experience. The original firmware on board was too old to support the new linux image booting. Kernel panics while booting. I spend three days succeeding. Attempted different combinations with HSS and Image. Discussing topics about PolarFireSoC are few online. Therefore, I made this document to guide newbies.

1. Program the latest HSS(1st stage bootloader) to PolarFire SoC.

(1). Download Official job file. **MPFS_ICICLE_KIT_BASE_DESIGN_ES_2026_04.tar**
<https://github.com/polarfire-soc/icicle-kit-reference-design/releases>

github.com/polarfire-soc/icicle-kit-reference-design/releases

Release list

- v2026.04
- v2025.07
- v2025.03
- v2024.09
- v2024.06
- v2024.02
- v2023.06
- v2023.02
- v2022.10
- v2022.10-rc1

[< Previous](#) [Next >](#)

v2026.04 Latest Compare

@v vfanis released this May 5 v2026.04 fd0aebb

Icicle Kit Reference Design Release v2026.04

Changes since last release

New Features:

- Connect pin 26 / RPI_GPIO7_SPI_CE1_N signal to GPIO2[1]
- Add AUTO_UPDATE and GOLDEN_IMAGE arguments for IAP programming
- Add counter to track MSS resets
- Use macro location references for CCC clock placement to improve clarity
- Add git hash information to design names to improve version tracking
- Add RPI header pin connection table to the README

Fixes and Improvements:

- Remove redundant DMA IRQ pin slicing causing intermittent build failures
- Update to support the latest version of libero 2025.2

Pre-built programming files

The [MPFS_ICICLE_KIT_BASE_DESIGN_2026_04.tar](#) (for MPFS250T) and [MPFS_ICICLE_KIT_BASE_DESIGN_ES_2026_04.tar](#) (for MPFS250T_ES) archives contain FlashPro Express programming files for the supported design variants.

The BASE_DESIGN_FD0AEBB95776A726D3 and BASE_DESIGN_ES_FD0AEBB95776A72 FlashPro Express programming files included in the tar files contain both the Libero FPGA design and the Hart Software Services v2026.04.1 release. These programming files can be used to program an Icicle Kit's FPGA and eNVM without running the Libero FPGA flow or building the HSS.

(2). Decompress the file and extract the job file.

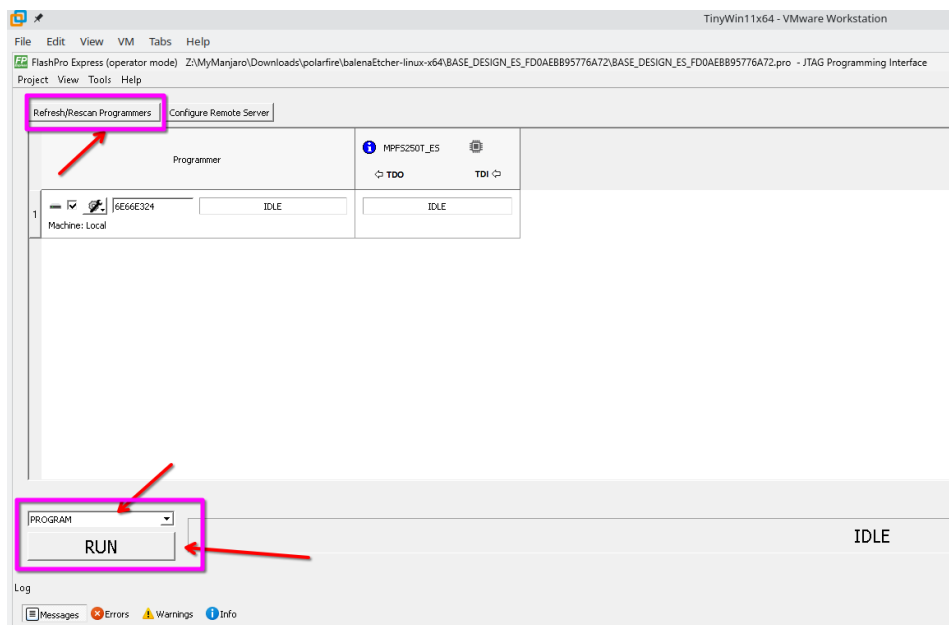
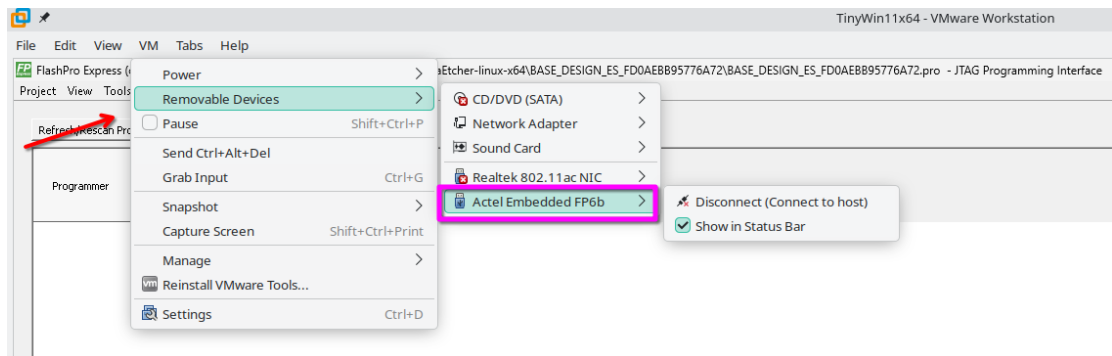
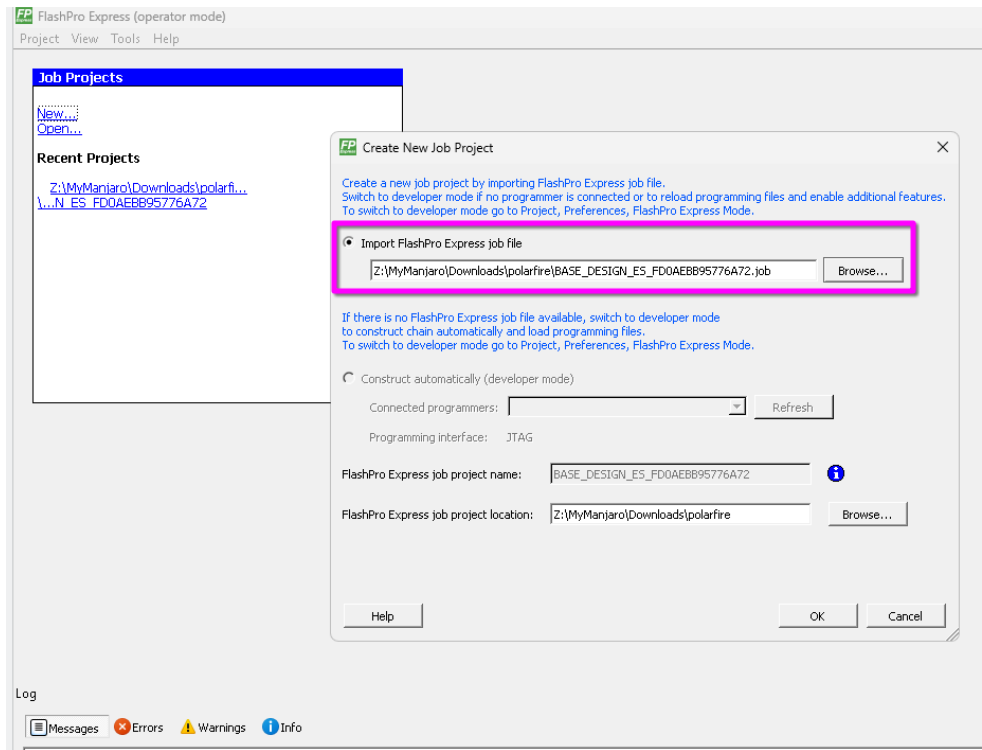
MPFS_ICICLE_KIT_BASE_DESIGN_ES_2026_04.tar — Ark

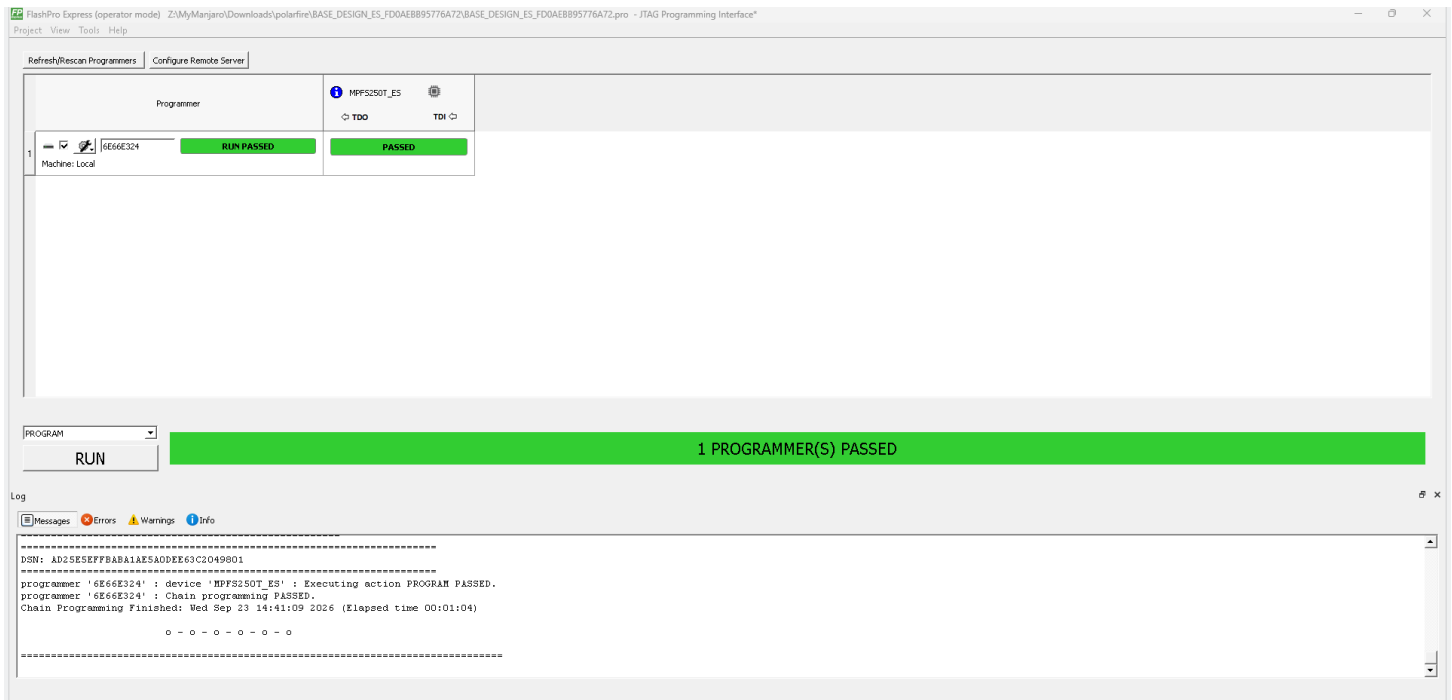
Extract Add Files Find Preview Open in External Application Remove from Archive

Name	Original Size	Mode	Owner	Group	Date
MPFS_ICICLE_KIT_BASE_DESIGN_ES_2026_04	11.7 MiB	drwxr-xr-x	bbb	bbb	4/30/26 11:49 PM
BASE_DESIGN_ES_FD0AEBB95776A72.job	11.7 MiB	-rw-r--r--	bbb	bbb	4/30/26 11:47 PM
BASE_DESIGN_ES_FD0AEBB95776A72.job.digest	467 B	-rw-r--r--	bbb	bbb	4/30/26 11:47 PM

MPFS_ICICLE_KIT_BA...

(3). Connect USB cable (J33) to Windows Operating System USB port.



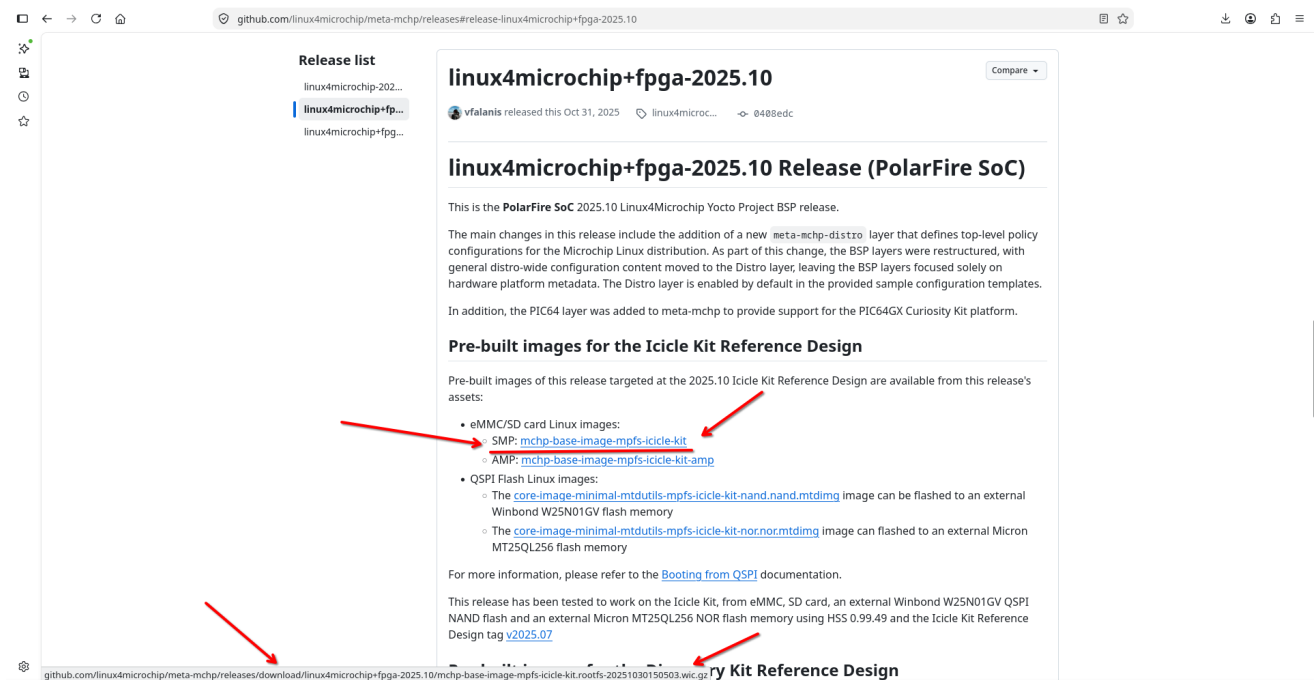


2. Program the latest linux system (u-boot+kernel+filesystem).

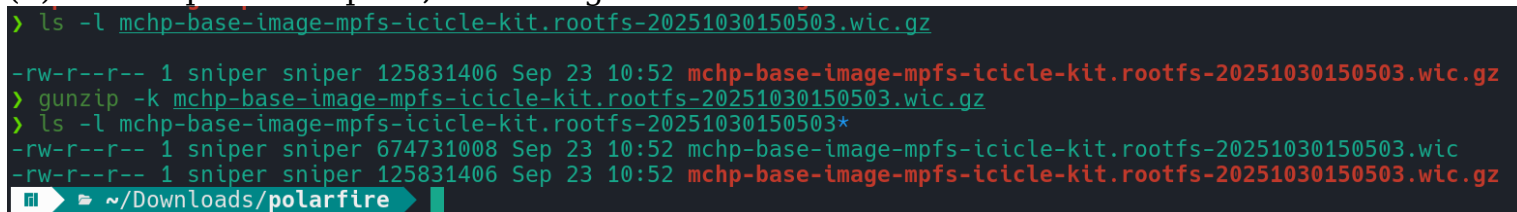
(1). Download the latest image.

mchp-base-image-mpfs-icicle-kit.rootfs-20251030150503.wic.gz

<https://github.com/linux4microchip/meta-mchp/releases>



(2). Decompress the pack, remove .gz suffix.



(3). Power ON the board, press any key to interrupt HSS starting up.

```

> sudo chmod 777 /dev/ttyUSB*
[sudo] password for sniper:
> ls -l /dev/ttyUSB*
crwxrwxrwx 1 root uucp 188, 0 Sep 23 15:23 /dev/ttyUSB0
crwxrwxrwx 1 root uucp 188, 1 Sep 23 15:23 /dev/ttyUSB1
crwxrwxrwx 1 root uucp 188, 2 Sep 23 15:23 /dev/ttyUSB2
crwxrwxrwx 1 root uucp 188, 3 Sep 23 15:23 /dev/ttyUSB3
~/Downloads/polarfire
>
~/Downloads/polarfire minicom -D /dev/ttyUSB0 -b 115200 -o
```

```

HDisk: zsh x polarfire: minicom x polarfire: minicom x polarfire: zsh x
Cached: SEG0_0: offset 0x0080000000, physical DDR 0x00000000
Cached: SEG0_1: offset 0x1000000000, physical DDR 0x00000000
Non-cached: SEG1_2: offset 0x00c0000000, physical DDR 0x00000000
Non-cached: SEG1_3: offset 0x1400000000, physical DDR 0x00000000
Non-cached WCB: SEG1_4: offset 0x00d0000000, physical DDR 0x00000000
Non-cached WCB: SEG1_5: offset 0x1800000000, physical DDR 0x00000000
[0.872534] L2 Cache Configuration:
L2-Scratchpad: 4 ways (512 KiB)
L2-Cache: 8 ways (1024 KiB)
L2-LIM: 4 ways (512 KiB)
[0.888479] DESIGNID: BASE_DESIGN_ES_FD0AEBB95776A72
[0.895068] DESIGNVER: 26.04
[0.899364] BACKLEVEL: 0000
[0.903565] Please ensure that jumpers J34/J43 are correct for 1.8V MMC voltage...
[0.913113] startup_service :: [init] -> [boot]
[0.919224] ipi_poll_service :: [Init] -> [Monitoring]
[0.926003] startup_service :: [boot] -> [idle]
[3.273657] usbdmcservice :: [Init] -> [Idle]
Press a key to enter CLI, ESC to skip
Timeout in 1 second
.[3.301910] Character 32 pressed
[3.306493] Type HELP for list of commands
>> help
Supported commands:
BOOT RESET HELP VERSION UPTIME DEBUG MEMTEST QSPI EMMC MMC SDCARD PAYLOAD SPI USBDMSC RESUME ECC
>>
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB0
```

(4). Type command `mmc` and then `usbdmhc` to expose eMMC as a USB storage device to Linux Host.

```
>>
Supported commands:
BOOT RESET HELP VERSION UPTIME DEBUG MEMTEST QSPI EMMC MMC SDCARD PAYLOAD SPI USBDMSC RESUME ECC
>> mmc
[117.476443] Selecting SDCARD/MMC (fallback) as boot source ...
>> usbdmhc
[123.476302] usbdmhc_service :: [Idle] -> [WaitForUSBHost]
>> [123.483845] initialize MMC
[123.487855] Attempting to select SDCARD ... Failed
[123.545230] Attempting to select eMMC ... Passed
[123.893983] MMC - 512 byte pages, 512 byte blocks, 15273600 pages
Waiting for USB Host to connect... (CTRL-C to quit)
. 0 bytes written, 0 bytes read
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB0
```

(5). Move USB cable from J11 to J16. Hotplug is okay, a new USB device is detected by Linux Host.

Disk /dev/sdb: 7.28 GiB, 7820083200 bytes, 15273600 sectors
Disk model: PolarFireSoC_msd
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 00000000-0000-0000-0000-00004D9B9EF0

Device	Start	End	Sectors	Size	Type
/dev/sdb1	8192	139263	131072	64M	Microsoft basic data
/dev/sdb2	139264	155647	16384	8M	BIOS boot
/dev/sdb3	155648	15273559	15117912	7.2G	Linux filesystem

~/Downloads/polarfire

Network

Recent

- Recent Files
- Recent Locations

Devices

- 87.1 GiB Internal Drive (nvme0n1p3)
- 107.4 GiB Internal Drive (nvme0n1p5)
- 931.5 GiB Internal Drive (sda1)

Removable Devices

- boot
- root

(6). Use `dd` command to flush image to eMMC.

```
> sudo dd if=mchp-base-image-mpfs-icicle-kit.rootfs-20260430113402.wic of=/dev/sdb bs=1M conv=fsync status=progress
79691776 bytes (80 MB, 76 MiB) copied, 6 s, 13.1 MB/s
91226112 bytes (91 MB, 87 MiB) copied, 7 s, 13.0 MB/s
645922816 bytes (646 MB, 616 MiB) copied, 51 s, 12.7 MB/s653196288 bytes (653 MB, 623 MiB) copied, 51.6486 s, 12.6 MB/s
622+1 records in
622+1 records out
653196288 bytes (653 MB, 623 MiB) copied, 51.951 s, 12.6 MB/s
~/Downloads/polarfire 52s
```

(7). Re-power on the board, open two minicom terminals, one for HSS(1st stage bootloader), another for uboot(2nd stage bootloader)+linux.

```

[4.910440] boot_service(u54_3) :: [SetupPMPCComplete] -> [ZeroInit]
[4.918460] boot_service(u54_4) :: [SetupPMPCComplete] -> [ZeroInit]
[4.926480] boot_service(u54_1) :: [ZeroInit] -> [Download]
[4.933737] boot_service(u54_2) :: [ZeroInit] -> [Download]
[4.940993] boot_service(u54_3) :: [ZeroInit] -> [Download]
[4.948250] boot_service(u54_4) :: [ZeroInit] -> [Download]
[4.955506] boot_service(u54_1)::Processing boot image: "u-boot.bin"
[4.963336] boot_service(u54_2) :: [Download] -> [Complete]
[4.970592] boot_service(u54_3) :: [Download] -> [Complete]
[4.977849] boot_service(u54_4) :: [Download] -> [Complete]
[5.00625] SYSREG:MAINTENANCE_INT_SR changed since last read (0xf080)
[5.09886] SYSREG:DLL_STATUS_SR changed since last read (0xf0017)
[5.85550] boot_service(u54_1) :: [Download] -> [OpenSBIInit]
[5.92997] boot_service(u54_1)::Registering domain "u-boot.bin" (hart mask 0x1e)
[5.101973] boot_service(u54_1)::Using built-in DTB at 0xa025b88
[5.109707] boot_service(u54_1) :: [OpenSBIInit] -> [Wait]
[5.118196] boot_service(u54_1) :: [Wait] -> [Complete]
[5.164573] u54 State Change: [SBIHartInit] [Booting] [Booting] [Booting]
[5.215464] boot_service(u54_1) :: [Complete] -> [Idle]
[5.224821] boot_service(u54_2) :: [Complete] -> [Idle]
[5.244073] boot_service(u54_3) :: [Complete] -> [Idle]
[5.263342] boot_service(u54_4) :: [Complete] -> [Idle]
[5.364326] u54 State Change: [Running] [SBIWaitForColdboot] [SBIWaitForColdboot] [SBIWaitForColdboot]
[12.297302] u54 State Change: [Running] [Running] [Running] [Running]
[51.117326] loop 2500000 took 12288 ticks (max 606580536 ticks)
[102.594378] loop 5000000 took 12287 ticks (max 606580536 ticks)

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB0

```

```
Hidisk: zsh x polarfire: minicom x polarfire: zsh x
[ OK ] Finished Resets System Activity Logs.
[ OK ] Reached target Preparation for Network.
Starting Network Configuration...
[ OK ] Started Avahi mDNS/DNS-SD Stack.
[ OK ] Started User Login Management.
[ OK ] Started Network Configuration.
[ OK ] Reached target Multi-User System.
t-User System 15.475312] macb 20110000.ethernet eth0: PHY [20112000.ethernet-ffffffff:08] driver [Generic PHY] (irq=POLL)
m.
[ 15.485189] macb 20110000.ethernet: gem-ptp-timer ptp clock registered.
[ OK ] Reached target Network.
[ 15.512942] macb 20112000.ethernet end0: PHY [20112000.ethernet-ffffffff:09] driver [Generic PHY] (irq=POLL)
[ 15.523608] macb 20112000.ethernet end0: configuring for phy/sgmii link mode
[ 15.532933] macb 20112000.ethernet: gem-ptp-timer ptp clock registered.
Starting Record Runlevel Change in UTMP...
[ OK ] Finished Record Runlevel Change in UTMP.
[ 19.598092] macb 20110000.ethernet eth0: Link is Up - 1Gbps/Full - flow control off

Microchip Distro 1.0 mpfs-icicle-kit ttyS1

mpfs-icicle-kit login: root
Your password has expired. Choose a new password.
Changing password for root
Enter the new password (minimum of 5 characters)
Please use a combination of upper and lower case letters and numbers.
New password:
Bad password: too short.
Warning: weak password (enter it again to use it anyway).
New password:
Re-enter new password:
passwd: password changed.
This is version v2026.04 of the meta-mchp release.

Updated images and documentation are available at:
https://github.com/polarfire-soc/
root@mpfs-icicle-kit:~# ls
root@mpfs-icicle-kit:~#
```

```
sudo
minicom -D /dev/ttyUSB1
-b 115200 -o

username: root
password: 12345678
```

```
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB1
```


(8). Configure IP address and ping Linux Host to check the network connection.

```
HDisk : zsh × polarfire : minicom × polarfire : minicom × polarfire : zsh ×
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 88 bytes 7624 (7.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 88 bytes 7624 (7.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

sit0: flags=128<NOARP> mtu 1480
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 1000 (UNSPEC)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mpfs-icicle-kit:~# ifconfig eth0 192.168.100.33
root@mpfs-icicle-kit:~# ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.100.33 netmask 255.255.255.0 broadcast 192.168.100.255
    inet6 fe80::204:a3ff:fe04:9802 prefixlen 64 scopeid 0x20<link>
    ether 00:04:a3:04:98:02 txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 628 (628.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 35 bytes 7096 (6.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 86

root@mpfs-icicle-kit:~# ping 192.168.100.211
PING 192.168.100.211 (192.168.100.211) 56(84) bytes of data.
64 bytes from 192.168.100.211: icmp_seq=1 ttl=64 time=0.583 ms
64 bytes from 192.168.100.211: icmp_seq=2 ttl=64 time=0.244 ms
64 bytes from 192.168.100.211: icmp_seq=3 ttl=64 time=0.352 ms
64 bytes from 192.168.100.211: icmp_seq=4 ttl=64 time=0.372 ms
64 bytes from 192.168.100.211: icmp_seq=5 ttl=64 time=0.240 ms
64 bytes from 192.168.100.211: icmp_seq=6 ttl=64 time=0.333 ms
64 bytes from 192.168.100.211: icmp_seq=7 ttl=64 time=0.358 ms

root@mpfs-icicle-kit:~# ping -I end0 192.168.100.211
PING 192.168.100.211 (192.168.100.211) from 192.168.100.66 end0: 56(84) bytes of data.
64 bytes from 192.168.100.211: icmp_seq=1 ttl=64 time=0.585 ms
64 bytes from 192.168.100.211: icmp_seq=2 ttl=64 time=0.239 ms

64 bytes from 192.168.100.211: icmp_seq=3 ttl=64 time=0.227 ms
64 bytes from 192.168.100.211: icmp_seq=4 ttl=64 time=0.219 ms
64 bytes from 192.168.100.211: icmp_seq=5 ttl=64 time=0.223 ms
64 bytes from 192.168.100.211: icmp_seq=6 ttl=64 time=0.215 ms
64 bytes from 192.168.100.211: icmp_seq=7 ttl=64 time=0.220 ms
64 bytes from 192.168.100.211: icmp_seq=8 ttl=64 time=0.268 ms

```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB1

September 23, 2026
Anonymous